

```
In [ ]: #Python - Functions
```

```
In [3]: def am9():  
        print('Good morning Team')
```

```
In [4]: def am9():  
        print('Good morning Team')  
        am9()
```

Good morning Team

```
In [5]: def greet():  
        print('Hello')  
        print('Good Morning')
```

```
In [6]: def greet():  
        print('Hello')  
        print('Good Morning')  
        greet()
```

Hello

Good Morning

```
In [7]: def greet():  
        print('Hello')  
        print('Good Morning')  
        greet()  
        def greet():  
            print('Hello')  
            print('Good Morning')  
            greet()  
        def greet():  
            print('Hello')  
            print('Good Morning')  
            greet()
```

```
Hello
Good Morning
Hello
Good Morning
Hello
Good Morning
```

```
In [8]: def greet():
        print('Hello')
        print('Good Morning')
        greet()
        greet()
        greet()
```

```
Hello
Good Morning
Hello
Good Morning
Hello
Good Morning
```

```
In [10]: def add(x,y):
        c = x+y
        print(c)
        add(5,6)
```

11

```
In [11]: def add(x,y):
        c = x+y
        print(c)
        add(5)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[11], line 4
      2         c = x+y
      3         print(c)
----> 4 add(5)

TypeError: add() missing 1 required positional argument: 'y'
```

```
In [12]: def add(x,y):  
         c = x+y  
         print(c)  
         add(5,6,7)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[12], line 4  
      2     c = x+y  
      3     print(c)  
----> 4 add(5,6,7)  
  
TypeError: add() takes 2 positional arguments but 3 were given
```

```
In [13]: def add(x,y):  
         c = x+y  
         print(c)  
         add(5,6,7,8)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[13], line 4  
      2     c = x+y  
      3     print(c)  
----> 4 add(5,6,7,8)  
  
TypeError: add() takes 2 positional arguments but 4 were given
```

```
In [14]: def add(x,y,z):  
         c = x+y+z  
         print(c)  
         add(5,6,7)
```

18

```
In [15]: def add(x,y,z):  
         c = x+y+z+m  
         print(c)  
         add(5,6,7)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[15], line 4
      2     c = x+y+z+m
      3     print(c)
----> 4 add(5,6,7)

Cell In[15], line 2, in add(x, y, z)
      1 def add(x,y,z):
----> 2     c = x+y+z+m
      3     print(c)

NameError: name 'm' is not defined
```

```
In [16]: def greet():
        print('Hello')
        print('Good Morning')
        greet()

        def add(x,y):
            c = x+y
            print(c)
        add(5,6)
```

```
Hello
Good Morning
11
```

```
In [17]: def greet():
        print('Hello')
        print('Good Morning')

        def add(x,y):
            c = x+y
            print(c)

        greet()
        add(5,6)
```

```
Hello
Good Morning
11
```

```
In [18]: def add(x,y):  
         c = x+y  
         print(c)  
  
         def sub(x,y):  
             d = x-y  
             print(d)  
  
         add(5,4)  
         sub(10,2)
```

9
8

```
In [20]: def add_sub(x,y):  
         c = x+y  
         d = x-y  
         print(c)  
         print(d)  
  
         add_sub(10,5)
```

15
5

```
In [21]: def add_sub(x,y):  
         c = x+y  
         d = x-y  
         return c,d  
  
         add_sub(10,5)
```

Out[21]: (15, 5)

```
In [23]: def add_sub(x,y):  
         c = x+y  
         d = x-y  
         return c,d  
  
         result1, result2 = add_sub(10,5)  
         print(result1, result2)
```

15 5

```
In [24]: def add_sub(x,y):  
         c = x+y  
         d = x-y  
         return c,d  
  
result = add_sub(10,5)  
print(result)
```

(15, 5)

```
In [25]: def evenodd(x):  
         if(x%2 == 0):  
             return "Even"  
         else:  
             return "Odd"  
print(evenodd(16))  
print(evenodd(7))
```

Even

Odd

```
In [26]: def person(name, age):  
         print(name)  
         print(age)  
person('nit',22)
```

nit

22

```
In [27]: def person(name, age):  
         print(name)  
         print(age)  
person(23)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[27], line 4
      2     print(name)
      3     print(age)
----> 4 person(23)

TypeError: person() missing 1 required positional argument: 'age'
```

```
In [28]: def person(name, age):
          print(name)
          print(age)
          person('nit')
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[28], line 4
      2     print(name)
      3     print(age)
----> 4 person('nit')
```

```
In [29]: def person(name, age):
          print(name)
          print(age)
          person('nit',22,23)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[29], line 4
      2     print(name)
      3     print(age)
----> 4 person('nit',22,23)

TypeError: person() takes 2 positional arguments but 3 were given
```

```
In [30]: def person(name, age):
          print(name)
          print(age)
          person(23,'nit')
```

```
23
nit
```

```
In [31]: def person(name, age):
          print(name)
          print(age+1)
          person(23, 'nit')
```

```
23
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[31], line 4
      2     print(name)
      3     print(age+1)
----> 4 person(23, 'nit')

Cell In[31], line 3, in person(name, age)
      1 def person(name, age):
      2     print(name)
----> 3     print(age+1)

TypeError: can only concatenate str (not "int") to str
```

```
In [32]: def person(name, age):
          print(name)
          print(age+1)
          person('nit', 22)
```

```
nit
23
```

```
In [33]: def person(name, age):
          print(name)
          print(age+1)
          person(age = 23, name = 'nit')
```

```
nit
24
```

```
In [34]: def person(name, age, city):
          print(name)
          print(age)
```



```
    print(city)
    person(age = 23, name = 'nit', city = 'Mumbai')
```

```
nit
23
Mumbai
```

```
In [37]: def person(name, age = 18):
          print(name)
          print(age)
          person('nit')
```

```
nit
18
```

```
In [38]: def person(name, age = 18):
          print(name)
          print(age)
          person('nit',23)
```

```
nit
23
```

```
In [ ]: #2-Dec-2025
```

```
In [ ]: #Variable Length Argument
```

```
In [39]: def sum(a,b):
          c = a+b
          print(c)
          sum(5,6)
```

```
11
```

```
In [40]: def sum(a,b):
          c = a+b
          return c
          sum(5,6)
```

```
Out[40]: 11
```

```
In [41]: def sum(a,b):
          c = a+b
```

```
    return c
sum(5,6,7,8,9,10)
```

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[41], line 4
      2     c = a+b
      3     return c
----> 4 sum(5,6,7,8,9,10)

TypeError: sum() takes 2 positional arguments but 6 were given
```

```
In [42]: def sum(a,*b):
          c = a+b
          return c
          sum(5,6,7,8,9,10)
```

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[42], line 4
      2     c = a+b
      3     return c
----> 4 sum(5,6,7,8,9,10)

Cell In[42], line 2, in sum(a, *b)
      1 def sum(a,*b):
----> 2     c = a+b
      3     return c

TypeError: unsupported operand type(s) for +: 'int' and 'tuple'
```

```
In [43]: def sum(a,*b):
          print(type(a))
          print(type(b))
          sum(5,6,7,8)
```

```
<class 'int'>
<class 'tuple'>
```

```
In [44]: def sum(a,*b):
          c = a

          for i in b:
```

```
    c = c+i
    print(c)
sum(5,6,7,8,9,10)
```

45

```
In [ ]: #Keyword Variable Length Arguments (kwargs)
```

```
In [45]: def person(name, *data):
          print(name)
          print(data)
          person('ALEX', 36, 'JOHN', 987767)
```

ALEX
(36, 'JOHN', 987767)

```
In [46]: def person(name, *data):
          print('name')
          print(data)
          person('ALEX', age = 36, home_place = 'Southcity', mob = 987767)
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[46], line 4
      2     print('name')
      3     print(data)
----> 4 person('ALEX', age = 36, home_place = 'Southcity', mob = 987767)

TypeError: person() got an unexpected keyword argument 'age'
```

```
In [47]: def person(name, **data):
          print('name')
          print(data)
          person('ALEX', age = 36, home_place = 'Southcity', mob = 987767)
```

name
{'age': 36, 'home_place': 'Southcity', 'mob': 987767}

```
In [ ]: #Global Variable vs Local Variable:
```

```
In [48]: a = 10
          print(a)
```

10

```
In [50]: a = 10
def something():
    b = 15
    print('in function',b)
    print('out function',a)
```

```
In [51]: a = 10
def something():
    b = 15
    print('in function',b)
    print('out function',a)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[51], line 4
      2 def something():
      3     b = 15
----> 4 print('in function',b)
      5 print('out function',a)

NameError: name 'b' is not defined
```

```
In [52]: a = 10
def something():
    b = 15
    print('in function',b)
    print('out function',a)
```

out function 10

```
In [53]: a = 10
def something():
    b = 15
    print('in function',b)
    something()
    print('out function',a)
```

in function 15
out function 10

```
In [54]: a = 10
def something():
    a = 55
    print('in function',a)
something()
print('out function',a)
```

in function 55
out function 10

```
In [57]: a = 10
def something():
    global a
    b = 15
    print('in function',b)
    print('Global variable',a)
something()
print('out function',a)
```

in function 15
Global variable 10
out function 10

```
In [58]: x = 10
def update_x():
    global x
    x+=5
update_x()
print(x)
```

15

```
In [59]: x = 10
def update_x():
    globals()['x'] += 5
update_x()
print(x)
```

15

```
In [ ]: #Program to count even and odd numbers
```

```
In [63]: def count(lst):
    even = 0
    odd = 0
    for i in lst:
        if i%2 ==0:
            even+=1
        else:
            odd+=1
    return even, odd
lst = [1,2,3,4,8,9,10]
even, odd = count(lst)
print("Even Number: {} and odd Number : {}".format(even,odd))
```

Even Number: 4 and odd Number : 3

```
In [ ]: #Fibonacci series
```

```
In [65]: def fib(n):
    a = 0
    b = 1
    print(a)
    print(b)
    for i in range(0,n):
        c = a + b
        a = b
        b = c
        print(c)
    fib(10)
```

0
1
1
2
3
5
8
13
21
34
55
89

